

Samuel Babichenko

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Quantitative researcher (PhD, statistics and applied probability) who translates ambiguous business and research questions into tractable models, evaluation metrics and production-oriented analyses. Research on dynamic games with strategic information; industry model development and validation on a \$200B consumer-lending portfolio.

SKILLS

Modeling	forecasting and backtesting, model validation, time series, Kalman filtering, Bayesian models, false-discovery control, Monte Carlo, stochastic control
ML and data	PyTorch, transformers, diffusion models, gradient boosting (XGBoost, LightGBM), NumPy/SciPy, Pandas/Polars
Engineering	Python, C++, CUDA, SQL; AI-assisted development (coding agents)

EXPERIENCE

Wells Fargo · Quantitative Researcher, Rotational Program · Charlotte, NC Jul 2026 – Present

Wells Fargo · Quantitative Researcher Intern, Mortgage Model Development · Charlotte, NC Jun – Aug 2025

- Built data-validation and backtesting pipelines in SQL and Python over large loan-level and time-series datasets, separating systematic forecast error from noise; identified persistent model-error signals that informed retraining cadence and monitoring.
- Led cross-functional work on prepayment forecasting for a \$200B mortgage portfolio, turning model-performance concerns into diagnostic metrics, backtests and recalibration analyses behind a model upgrade projected to add substantial annual profit.
- Designed an improved cohorting method that accelerated model runtime, and quantified the accuracy–speed tradeoff with diagnostic tools that let non-technical stakeholders interpret model output.

UC Santa Barbara · PhD Researcher, Statistics and Applied Probability · Santa Barbara, CA Sep 2021 – Jun 2026

- Developed an exact solution method for multi-agent prediction games with private signals, in which players learn from each other's actions and can shape what others believe: equilibrium strategies and welfare costs without approximation or mean-field limits. Solo-authored paper (arXiv: 2603.12140), six-chapter dissertation.
- Quantified how information asymmetry distorts aggregate forecasts: systematic mean bias, excess volatility and wasted effort from agents optimizing over each other's predictions; showed that sharper private information can raise total cost. Presented at the Southern California Quantitative Finance Forum.
- Passed all qualifying exams before the first term; taught 12 courses, from introductory statistics to stochastic processes and financial mathematics; mentored 10+ undergraduate researchers, most now in PhD programs.

PROJECTS

noisestate · open-source solver, Python · pypi.org/project/noisestate

- Designed, built and published as sole author an open-source package on PyPI: users write a model as equations and get optimal strategies, costs and responses to shocks.
- Tells users when an answer can't be trusted; an interactive browser version lets anyone write and solve their own models (sbabichenko.com/noisestate).

Decision Mesh · C++17, Python · github.com/sbabichenko/triangular-decision-mesh

- Adaptive, tree-inspired regression that fits a smooth prepayment surface to public Ginnie Mae pool data, refining only where a false-discovery-rate test finds signal; separates the surface shared by all pools from each pool's own effect and noise.

Multi-agent trajectory generation · PyTorch · Waymo Open Motion Dataset

- Reimplemented BehaviorGPT, a transformer for multi-agent traffic simulation, and built an evaluation framework for closed-loop simulation (collisions, off-road events, failures at yielding, lane changes and intersections); extending it with a conditional diffusion model at multimodal branch points, targeting the core weakness of uniform-rate autoregression.

EDUCATION

UC Santa Barbara · PhD, Statistics and Applied Probability (MA 2026) 2026

UC San Diego · BS, Mathematics · Dean's Undergraduate Excellence Award (top 0.5%, Physical Sciences) 2021

Relevant coursework: reinforcement learning, neural networks for control, optimization, stochastic control, Monte Carlo methods, numerical analysis, parallel computing, econometrics